Spring 2025 MAT5210 Midterm Examination

1

. Let K be a field.

(a) What does it mean for a field extension $L : K$ to be normal?

(b) Give an example of field extensions $M \supseteq L \supseteq K$ such that both $M : L$ and $L : K$ are normal, but $M : K$ is not normal.

2

. Let $L : K$ be a finite field extension.

(a) What does it mean for $L : K$ to be Galois? State the Fundamental Theorem of Galois Theory.

(b) Suppose $L : K$ is Galois with $\mathrm{Gal}(L/K) \cong S_5$. How many elements in $\mathrm{Gal}(L/K)$ are of order 5?

(c) Prove that there are precisely six intermediate fields $M_1, M_2, \ldots, M_6$ between $L : K$ such that $[L : M_i] = 5$. And prove that none of the extensions $M_i : K$ is Galois.

3

. Let $L = \mathbb{Q}(\sqrt{2}, \sqrt[3]{3}, \sqrt{5})$.

(a) What is $[L : \mathbb{Q}]$? Give a basis for L over $\mathbb{Q}$. (You need not prove that your answer is correct.)

(b) Find all elements in $\mathrm{Aut}_{\mathbb{Q}}(L)$. Hence show that $L : \mathbb{Q}$ is Galois.

(c) Which well-known group is $\mathrm{Gal}(L/\mathbb{Q}) = \mathrm{Aut}_{\mathbb{Q}}(L)$ isomorphic to?

(d) For each of the following fields K_i, determine the subgroup $H_i \leq \mathrm{Gal}(L/\mathbb{Q})$ that corresponds to K_i under the Galois correspondence: $K_1 = \mathbb{Q}(\sqrt{10})$, $K_2 = \mathbb{Q}(\sqrt{6}, \sqrt{15})$, $K_3 = \mathbb{Q}(\sqrt{2} + \sqrt{5})$, $K_4 = \mathbb{Q}(\sqrt{30})$.

(e) Use the Galois correspondence (in Question 1(a)) to show that $K_1 \leq K_3$, then prove the same thing by a direct calculation.

(f) How many fields M are there with $\mathbb{Q} \subseteq M \subseteq L$ and $[M : \mathbb{Q}] = 4$?

(g) Show that if $f(x) \in \mathbb{Q}[x]$ is an irreducible monic polynomial of degree 3, then $f(x)$ has no roots in L.